

Annex VIII — Perceived Real-Time Visual Information Delivery

BEAP™ / WR Desk™ / Optirando™ Canonical Specification — standalone Annex. Version 1.0 (see §VIII.13).

Document status: Normative unless explicitly marked as informative.

Applies to: WR Desk™ runtime, the Host user interface and Presentation Surfaces (Annex V), the Augmented Overlay layer, the Night Pipeline (Annex III), the Forward Preparation lifecycle in both modes (Annex VI), the Morning Digest and Preparation Brief presentation, Event Sessions (Annex VI §VI.5), and all visual information artifacts composed for presentation to the operator.

Trademark notice: Optirando™, WR Desk™, WR Graph™, BEAP™, PoAE™, PoAC™, WRVault™, and WR Code® are used throughout as product and protocol designations of the Optirando™ system.

VIII.0 Status, Scope, and Relationship to the Canon

This Annex specifies **Hybrid Composition**: the mechanism by which visually rich, provenance-complete information artifacts — infographics in particular — are delivered to the operator on ordinary local hardware at perceived real-time, by splitting the work along its natural cost structure. Heavy semantic reasoning is performed during the preparation phase, by the strongest permitted source; presentation-time work is reduced to deterministic composition of attested building blocks against a declarative description. It further specifies the **Prepared Edition**: the editorial visual presentation form of the Morning Digest and Preparation Briefs.

This Annex does not redefine wire formats, capsule serialization, cryptographic primitives, transport, PoAE™ execution semantics, the preparation lifecycle, or the media reconstruction model. It depends on, reuses, and does not weaken:

- **Annex II** — the non-pre-satisfaction rule (§II.2.5), routing-state exclusivity (§II.2.5), graceful degradation (§II.2.8), and Forward Preparation triggers (§II.6.5);
- **Annex III** — the mutation class taxonomy and its assignment-table extension rule (§4.5), evidence and Digest discipline (§5), the escalation ladder (§6.2), and the Cloud Abstraction Boundary (§7);
- **Annex IV** — LoRA-class adapter governance including Suggest Mode (§IV.3.3), payload classes and the two distribution paths (§IV.6.2–IV.6.3), and the mandatory evidence annotation (§IV.6.5);
- **Annex V** — reconstruction-based display as the universal presentation path; every Presentation Surface rule of Annex V applies to the artifacts of this Annex;
- **Annex VI** — the Forward Preparation lifecycle in both modes, the Preparation Run stages (§VI.3.2), the Preparation Brief and Deep-Dive Expansion (§VI.3.3–VI.3.4), the Preparation Queue (§VI.4), Event Sessions (§VI.5), the Debrief (§VI.6), and retrieval privacy (§VI.7);
- **Annex VII** — the artifact and manifest machinery (§6), the verification chain (§7), the grant model (§10), and the execution boundary (VII-10.1).

The core philosophy of the Canon applies without exception: *the system prepares, the human authorizes, the system records cryptographic proof*. Everything specified in this Annex is preparation and presentation. Nothing in it creates, implies, or pre-satisfies any execution authority (Annex II §II.2.5; VII-10.1).

VIII.1 Motivation (Informative)

Frontier-scale AI systems produce visual information artifacts of high semantic quality — a well-reasoned infographic condenses minutes of reading into seconds of comprehension — but at generation latencies of tens of seconds, because semantic reasoning and pixel synthesis run through the same generative machinery. This makes them unsuitable for the moments this system exists to serve: the Morning Digest that must be pleasant and instant, and the Event Session where the operator needs the answer now. Conversely, local models on ordinary hardware respond quickly but lack the reasoning depth for high-quality visual information design.

The resolution is the placement principle the corpus already applies to knowledge and computation (Annex IV §IV.5.1), extended to visual presentation: reasoning is placed where time is abundant and the strongest source is permitted to work — the preparation phase; composition is placed where latency matters — presentation time — and reduced to deterministic assembly of attested building blocks. Real-time graphics amortizes cost the same way, precomputing expensive results and reconstructing cheaply at runtime; but there, runtime reconstruction is permitted to be approximate. For information, approximation at presentation time is failure: a plausible but incorrect value displayed with visual authority is a defect of the highest severity class. The placement of the neural component is therefore inverted relative to the graphics precedent and fixed as an invariant (VIII.3.2): neural reasoning only in preparation, deterministic composition at presentation.

VIII.2 Definitions

Terms defined in Annexes II–VII apply as defined there; in particular **Forward Preparation**, **Predictive Forward Preparation**, **Declared Event**, **Preparation Consent**, **Preparation Run**, **Preparation Brief**, **Preparation Queue**, **Event Session**, **Debrief**, and **Morning Digest** (Annex VI §VI.2; Annex III §2). This Annex adds:

Composition Description (CD). A declarative, schema-constrained JSON document fully specifying a visual artifact: layout structure, the identity of every element, the data bound into each slot, and the provenance and confidence metadata of VIII.4.2. A CD is a structured textual artifact (text-purity preserved) and MAY be carried in BEAP™ capsules like any other text-pure content.

Composition Engine (CE). A locally resident, deterministic rendering component transforming a valid CD into a composed HTML/CSS artifact displayed on a Presentation Surface (Annex V), per the rendering-target rules of VIII.4.5. The CE performs no inference and holds no network capability. The CE is tool-governed: whitelisted with pinned identity and content hash, consistent with the converter-governance discipline of Annex V §V.6.

Publisher Composition Library (PCL). A set of publisher-produced artifacts — visual assets, layout grammars, composition exemplars, domain presentation policy, and media references —

published and acquired under the artifact machinery of Annex VII (§§6, 7, 13) and admitted as Acquired Capabilities under Annex III §3.8 (VIII.5).

Derived Composition Asset. A visual asset — icon, pictogram, illustrative graphic, or pre-composed chart — generated locally during the preparation phase by tool-governed local generation, anchored to the WR Graph™ context region of its topic, and governed as a Derived Capability with a full derivation chain (VIII.5.6). Derived Composition Assets are the locally produced counterpart of PCL assets; neither class receives elevated authority over the other.

Composition Grammar. The constraint set against which every CD SHALL validate: the intersection of the CE base schema and the constraints declared by the applicable admitted PCL(s).

Candidate Set. The output of branched candidate preparation (VIII.6.1): a set of parameterized CDs, each bound to a Branch Condition, covering the anticipated outcome space of a Declared Event.

Branch Condition. A machine-evaluable predicate over locally available runtime data, deciding deterministically — without inference — whether a candidate CD applies to the actual circumstance.

Session Composition Pool. A session-scoped, locally generated set of pre-validated, parameterized CDs covering the active session's topic space, maintained for low-latency scored retrieval during live interaction (VIII.6.5). Pool candidates are ephemeral presentation artifacts: derived, reconstructible, discarded or condensed at session end.

Fitness Function. A deterministic, reproducible scoring algorithm ranking Session Composition Pool candidates against the current information need (VIII.6.5.2).

Refinement Vocabulary. The closed, deterministic set of transformation operations a Composition Grammar permits on a pre-validated candidate at presentation time (VIII.6.5.3).

Augmented Overlay. An operator-toggable interactive presentation layer rendered over composed artifacts and Presentation Surfaces (VIII.9): annotations, context-bound automation suggestions, and spoken and cursor-based interaction affordances, composed from the same libraries and governed by the same disciplines as the underlying artifacts.

Prepared Edition. The editorial-form visual presentation of the Morning Digest and of Preparation Briefs (VIII.8): assembled during the preparation phase, presented for rapid ingest, with Deep-Dive Expansion per Annex VI §VI.3.4.

Trigger provenance. The mandatory CD attribute recording the preparation mode of the underlying content: **declared** (Forward Preparation — a confirmed Declared Event) or **inferred** (Predictive Forward Preparation — a Preparation Queue entry not yet accepted).

VIII.3 The Hybrid Compute Architecture

VIII.3.1 Compute Split (Normative)

The work of producing a visual information artifact is split as follows:

Phase	Component	Work	Governing rules
Preparation	External	Generic composition	Annex III §7 (Cloud Abstraction

Phase	Component	Work	Governing rules
	reasoning (cloud), where configured	candidates, layout reasoning, curriculum-level design exploration — at taxonomy level only	Boundary); Annex VI §VI.7 (masking, Obfuscation Mode); Preparation Consent scope (VI.3.1)
Preparation	Admitted LoRA-class adapters and PCL exemplars	Domain-specific composition competence, applied locally	Annex IV §IV.3 (Suggest Mode, context-scoped activation); VIII.5
Preparation	Local instantiation (mandatory)	Mapping generic candidates onto concrete local data, slots, and graph content	Annex III §7.4
Presentation	Local model	Bounded selection only: Branch Condition disambiguation, candidate identification, binding map (VIII.6.3)	Grammar-constrained decoding (VIII.4.1)
Presentation	Composition Engine	Slot instantiation, validation, deterministic rendering	VIII.4, VIII.7; Annex V presentation rules

Boundary compliance. External reasoning in service of composition is a preparation-phase external interaction and SHALL comply in full with the Cloud Abstraction Boundary (Annex III §7) and the retrieval-privacy discipline of Annex VI §VI.7. In particular: WR Graph™ content, event descriptors, and operator data SHALL NOT cross the boundary (Annex III §7.3; VI.7.3); external reasoning operates on masked, taxonomy-level requests and returns **generic** candidate structures; binding generic candidates to real data, real slots, and real graph content is the mandatory local instantiation step (Annex III §7.4). Hybrid Composition in its entirety SHALL be operable with external reasoning disabled — local-plus-PCL composition is always a conforming mode (Annex III §7.5; VI.7.2). External reasoning is an accelerant, never a dependency.

Informative — why hybrid rather than local-only or cloud-only: local-only composition is fast but semantically shallow without adapter and exemplar support; cloud-only generation is tens of seconds too slow and would place presentation-time context at an external party, which the boundary prohibits regardless. The hybrid split obtains strong reasoning where it is permitted to operate (generic, masked, prepared ahead), spends reasoning effort once per preparation instead of once per display, and keeps every concrete datum local.

VIII.3.2 The Inversion Rule (Normative)

Neural reasoning SHALL occur only in the preparation phase, where its output is validated and captured as provenance-bound artifacts. The presentation path SHALL be deterministic, with two bounded exceptions: bounded selection (VIII.6.3) and grammar-constrained generation on the staged fallback path (VIII.6.6, including presentation-time text generation per VIII.3.3).

A conforming implementation SHALL NOT introduce any presentation-time component whose output for identical inputs is not reproducible. Reproducibility of the presentation path is a precondition of the audit property (VIII.11.3).

VIII.3.3 The Composition-Not-Synthesis Rule (Normative)

Every visual element displayed under this Annex SHALL originate from exactly one of:

1. the local host asset base of the WR Desk™ runtime;
2. an admitted PCL (VIII.5) — for media-class assets, in their Reconstructed Artifact form per Annex V;
3. a Derived Composition Asset generated during the preparation phase under VIII.5.6;
4. operator-local data bound into slots: WR Graph™ content, admitted Sources, session evidence.

No visual element — imagery, icon, graphic, or any media-class asset — SHALL be synthesized at presentation time. **Textual content MAY be generated at presentation time:** grammar-constrained, validated before display (VIII.4.1, VIII.7(1)), with every data value within it bound from provenance-bound sources (graph content, admitted Sources, prepared material) — narrative text is generated, facts are only ever bound. This is the text counterpart of the staged fallback (VIII.6.6), where the Stage-1 textual core and attention-window generation already rely on it. Visual asset synthesis is permitted exclusively in the preparation phase, under the governance of VIII.5.6 — the inversion rule (VIII.3.2) applied to assets: what may not happen at the moment of display may happen ahead of it, validated, provenance-bound, and reversible. Adapter-shaped composition remains within Suggest Mode semantics (Annex IV §IV.3.3): parametric competence proposes *how content is composed*; the content values themselves are bound from the declarative and episodic dimensions, never asserted by the adapter. Consequently every element carries a resolvable provenance chain (VIII.4.3), and every composed artifact is exactly reproducible from its CD plus the referenced, hash-bound assets.

VIII.4 The Composition Description

VIII.4.1 General Requirements (Normative)

A CD SHALL: validate against the applicable Composition Grammar before any part of it is displayed; be text-pure — assets referenced by identifier and content hash, never embedded; and be self-contained — rendering SHALL NOT require inference, network access, or information beyond the CD, the referenced admitted assets, and the bound local data.

When produced by the local model, CD generation SHALL be grammar-constrained (schema-enforced decoding), making syntactically invalid output unrepresentable. When produced during a Preparation Run — including from external generic candidates after local instantiation — a CD SHALL be validated against the same grammar before staging; a CD failing validation SHALL be discarded, never repaired silently (consistent with the no-silent-error principle).

VIII.4.2 Mandatory Provenance and Confidence Metadata (Normative)

Every CD SHALL carry, at document level and inheritable per section:

- **trigger_provenance** ∈ { `declared`, `inferred` } per VIII.2;
- **confidence** — for `inferred` content, the evidence strength of the underlying prediction (Annex VI §VI.4); for `declared` content the field records `declared`, relevance being established by the confirmed event descriptor;
- **preparation_ref** — reference to the Preparation Consent and Preparation Run (for `declared`), or to the Preparation Queue entry and its evidence chain (for `inferred`).

These fields are not advisory: VIII.8.3 binds them normatively to visual salience.

VIII.4.3 Per-Element Provenance (Normative)

Every element within a CD SHALL declare its origin: `host, pcl:<publisher-id>:<artifact-id>@<hash>, derived:<asset-id>@<hash>` (VIII.5.6), or `local:<binding>`. The rendered artifact SHALL preserve the ability to resolve, for any displayed element, its origin and — for PCL elements — the PoAC™ acquisition record of the containing library, or — for Derived Composition Assets — the derivation chain and Digest decision state (Annex III §3.1, VIII.5.6.2–.3). This extends the provenance-display discipline of Annex IV §IV.3.5 and Annex V §V.4(2) to composed artifacts: composition influence is never silent.

VIII.4.4 Slots (Normative)

Templates declare typed slots with fixed geometry. Slot instantiation SHALL be a pure binding operation — insert current values, format per domain presentation policy (VIII.5.4), validate. Slot geometry is fixed at template level so that instantiation cannot alter layout (VIII.7, no layout shift).

VIII.4.5 Rendering Target — HTML/CSS (Normative)

The composed artifact produced by the CE is an **HTML/CSS document**: layout, typography, and styling are expressed exclusively in HTML markup and CSS. This target is chosen deliberately — it is itself a declarative, structured textual form (text-purity preserved for the document body), deterministically renderable, diffable, and inspectable, and its constraint-based layout model (grid, flow) is the constraint solver of VIII.8.2 in native form.

The following restrictions are mandatory:

1. **Strict passivity.** The composed artifact SHALL contain no script, no active content, no event handlers, and no form of executable or self-modifying behavior. Interactivity (Deep-Dive descent, consent affordances, media playback) is provided by the Presentation Surface around and over the artifact — never by the artifact itself. A composed artifact is display data in exactly the sense of Annex V: it is rendered, never executed.
2. **Library-only resources.** Every non-textual resource the document references — raster and vector imagery, icons, illustrative elements, fonts, and any other typographic or graphical asset — SHALL resolve exclusively to the local hash-addressed asset store: host assets or admitted PCL assets (for media-class assets, their Reconstructed Artifacts per VIII.5.1). References are by local content-hash address; the document SHALL contain no external URL, no network-resolvable reference, and no resource whose retrieval at render time would require connectivity. Rendering a composed artifact SHALL cause zero network activity.
3. **Deterministic subset.** The CE SHALL restrict output to a pinned, versioned HTML/CSS feature subset whose rendering is deterministic on the deployment's Presentation Surfaces; the subset definition is part of the CE's tool-governance identity (its version and hash, VIII.2). Features with environment-dependent or time-variant rendering behavior are excluded from the subset.
4. **Self-containment.** The document plus the referenced hash-addressed assets constitute the complete rendering input (VIII.4.1 self-containment, restated for the target format). The retained CD, the CE identity, and the asset set therefore reproduce the artifact exactly (VIII.11.3).

Informative. This resolves the media-class question for the composed artifact itself: the infographic as a whole is never an image. It is structured text over attested assets — which is why it can be validated before display, degraded gracefully (VIII.7(4)) — the degradation chain is a sequence of progressively simpler HTML documents over the same bindings), diffed between versions, and audited element by element, none of which a rasterized artifact permits.

VIII.5 Composition Libraries — Acquired and Derived

Composition libraries exist in two provenance classes, mirroring the capability model of Annex III §2: **Acquired** — the Publisher Composition Library (VIII.5.1–VIII.5.5) — and **Derived** — locally generated assets (VIII.5.6). Both are hash-addressed in the local asset store, both are resolvable by the CE, and neither carries elevated authority.

VIII.5.1 PCL — Artifact Model, Distribution, and Admission (Normative)

A PCL is a set of publisher artifacts under the machinery of Annex VII: each PCL artifact carries a signed manifest, is committed as a leaf under the publisher's Manifest Root, and SHALL pass the full verification chain of Annex VII §7 before staging. Distribution occurs over the two paths of Annex IV §IV.6.3 — public repository (publicly auditable) or direct message path under an established handshake, with the mandatory evidence annotation of Annex IV §IV.6.5. Delivery to a consumer occurs only under the grant model of Annex VII §10: PCL delivery and update are delivery/preparation rights; consistent with VII-10.1, nothing about a PCL confers or implies execution authority.

Admission of a PCL — in whole or per artifact — is a Class-3 acquisition under PoAC™ (Annex III §§3.8, 4.4–4.5), with the standard acquisition evidence record. Until admitted, PCL artifacts are unavailable to composition. Re-verification failure is an invalidated acquisition with cascade treatment (Annex III §3.5).

Repository hosting and the two paths. PCL artifacts — composition assets, layout grammars, exemplars, domain presentation policy, and overlay content (VIII.9.2) — are distributable units in the full sense of Annex VII §1 and follow the two distribution paths without exception or special case:

1. **Public path.** Library elements serving public offerings (pBEAP™) are hosted in the public Optirando™ repository (optirando.com) under the publisher's Manifest Root, alongside templates, LoRA-class adapters, and capability declarations. The public-auditability rule of Annex IV §IV.6.3(1) applies in full: every library element a public scan can deliver — every icon, grammar, exemplar, policy document, and overlay annotation set — SHALL be publicly inspectable and auditable, before and after delivery; no sealed content is hosted on the public path. What composes the operator's infographics is thereby exactly as open to inspection as what automates their workflows.
2. **Secret path.** Non-public library content follows Annex VII §13.2 unchanged: artifacts self-hosted, committed as leaves under the same signed per-provider Manifest Root (one root covers public and secret offerings alike; per-offering commitments remain prohibited — cardinality leaks offering structure), hash-bound, with evidentiary reveal against the timestamped public root, mandatory publisher attestation (VII-8.8), and the mandatory

evidence annotation of Annex IV §IV.6.5 on direct delivery. Secrecy hides library content, never its existence-proof and never publisher identity.

Publication-side reconstruction (Normative). The public repository SHALL NOT host media originals in any distributable or resolvable form. Originals MAY be retained on the repository side as **Custody Originals** in the sense of Annex V §§V.2, V.3.3: encrypted, inert, hash-bound blobs kept for deterministic evidence purposes — proving, byte-exactly, what a publisher submitted and from which original a published reconstruction derives. A custody-retained original SHALL NOT be resolvable by any scan, fetch, or delivery path; access to it is an exceptional, policy-gated, logged act (forensics, dispute resolution), never part of ordinary operation — the custody discipline of Annex V applied to the publication side. Publication of any media-class library asset (raster imagery, illustrative graphics, and any format subject to Annex V) requires its conversion, on the repository's infrastructure, by a tool-governed Trusted Converter (whitelisted, pinned identity and content hash, ephemeral converter instances — the converter-governance discipline of Annex V §V.6 applied to the publication side) into a Reconstructed Artifact carrying full Reconstruction Provenance (Annex V §V.3.3). **WebP is the pinned raster serialization for library imagery**, in the same sense in which safetensors is the pinned serialization for LoRA-class adapters (Annex IV §IV.3.2): submitted images are decoded to raw pixels and re-encoded to WebP by the repository's trusted encoder — fresh container, fresh encoding, none of the original byte structure, no binary metadata carry-over; potentially malicious originals are thereby never passed on. A change of the pinned raster format is a governance change of the converter identity, not a per-publisher option. What the repository hosts, what a scan delivers, and what any deployment fetches is exclusively the reconstruction: no original byte structure enters the distribution chain, and consequently no original can ever reach a host environment through it.

Secret-path reconstruction location. On the secret path, no repository infrastructure is involved: artifacts are self-hosted, and there is no publication-side conversion step. Media-class library assets delivered over the secret path are therefore handled exactly like any other directly delivered media, as previously specified: mandatory evidence annotation at dispatch (Annex IV §IV.6.5), and reconstruction performed by the **receiving deployment's sandbox pipeline** (Annex V §V.3) before any composition. The structural guarantee is identical on both paths — only its location differs: the public path reconstructs at publication and again at receipt; the secret path reconstructs at receipt.

Publication-side reconstruction is **additive, never a substitute**: it removes originals from the distribution chain at the source, but the receiving deployment's obligations are unchanged — transport confers no trust (Annex IV §IV.3.8), fetched media-class assets remain inbound media and traverse the receiving side's Annex V pipeline before the CE may compose them, and the full verification chain of Annex VII §7 gates staging regardless of path. The High-Assurance disciplines of Annex V apply to library assets identically to every other media class; a library element enjoys no relaxation for being small, decorative, or repository-hosted.

Media-class assets. PCL assets that are media or documents (raster icons, illustrative imagery) are inbound media in the sense of Annex V: they SHALL traverse the reconstruction pipeline, and the CE composes exclusively their Reconstructed Artifacts. Text-pure vector and structural assets remain structured textual artifacts. Blob-inertness holds without exception.

VIII.5.2 Library Contents (Normative)

A PCL MAY contain: **assets** (icons, pictograms, illustrative elements, themes, typographic resources); **layout grammars** (the permitted composition space: grids, section structures, template families with slot definitions, permitted element combinations); **exemplars** (curated, publisher-reviewed pairs of data pattern → complete CD); **domain presentation policy** (VIII.5.4); and **media references** (VIII.5.5).

VIII.5.3 Constraining the Composition Space (Normative)

The quality floor of local composition is governed by how tightly the composition space is constrained, not by the size of the local model. Accordingly:

- A publisher SHALL constrain the composition space via layout grammars. Free-form placement by the local model is non-conformant wherever a governing PCL is admitted.
- A publisher SHOULD supply exemplars covering its domain's recurring data patterns. Where exemplars exist, local generation SHOULD proceed by exemplar retrieval (via the WR Graph™) followed by slot-filling and bounded adaptation, in preference to free generation.
- Parametric composition competence (a LoRA-class adapter for composition style) MAY be published and acquired under Annex IV §IV.3 unchanged; on the remediation logic of the escalation ladder (Annex III §6.2), grammars and exemplars are the cheaper rungs and SHALL be exhausted before parametric competence is proposed.

VIII.5.4 Domain Presentation Policy (Normative)

Domain presentation policy is the publisher's machine-readable presentation rulebook: axis and unit conventions, color semantics, rounding rules, mandatory elements (source line, timestamp), forbidden encodings. It SHALL be applied as a deterministic post-validation step on every CD governed by the PCL. A violating CD SHALL be rejected, or deterministically corrected according to rules declared in the policy itself; probabilistic repair is not permitted.

VIII.5.5 Media References (Normative)

Explanatory video and audio SHALL be integrated by reference only: an attested identifier plus time-range from the signed PCL manifest, composed as a link element. Reconstruction and display of the referenced material follow Annex V unchanged. Retrieval of referenced media SHALL NOT be correlatable by the publisher with the occurrence of an Event Session or the content of any preparation (threat model of Annex VI §VI.7.1); implementations SHOULD bundle referenced media during the preparation window under the Preparation Consent rather than retrieve at presentation time, and where retrieval at presentation time occurs it is subject to the masking and Obfuscation Mode rules of Annex VI §VI.7 — in High-Assurance Mode, Obfuscation Mode is mandatory per VI.7.4.

VIII.5.6 Derived Composition Assets — Locally Generated Library Content (Normative)

A composition library need not originate with a publisher. During night and idle periods, the Night Pipeline MAY generate composition assets fitted to the deployment's active topics — icons, pictograms, illustrative graphics, and pre-composed charts over admitted graph data — and stage them, hash-addressed, anchored to the WR Graph™ context region of the topic they serve.

VIII.5.6.1 Generation discipline. Asset generation is a preparation-phase operation under the inversion rule (VIII.3.2). Generation tools are tool-governed exactly like the CE and the Trusted Converter: whitelisted, with pinned identity and content hash (Annex V §V.6 discipline); outputs are hash-bound at creation. Generation draws exclusively on admitted content: graph data (for charts), admitted Sources, existing admitted assets, and the generation tool's own capability. Where any external interaction serves generation, the Cloud Abstraction Boundary applies in full (Annex III §7): taxonomy-level requests only, mandatory local instantiation, and Class-3 consent for the interaction.

VIII.5.6.2 Provenance. Every Derived Composition Asset is a Derived Capability (Annex III §2) and SHALL carry a full derivation chain per Annex III §3.1: originating evidence (the topic-region signals that motivated it), the generation tool identity and version, the input entities it draws on, its content hash, and its target context region. As with derived adapters (Annex III §3.4), the durable asset is the derivation recipe: a Derived Composition Asset is regenerable by design, including after a generation-tool upgrade.

VIII.5.6.3 Staging, reporting, and retroactive rejection. Generation and staging of Derived Composition Assets from admitted content is a Class-2 operation: executed autonomously during the Night Pipeline, reported item-by-item in the Morning Digest with a one-action rollback, prior state retained for the configurable retention window (Annex III §4.3). Staged assets are available to composition immediately upon staging — this is the deliberate difference from the Class-3 admission of acquired PCL content, justified by provenance class: nothing external enters, nothing leaves, and everything is reversible. The operator's Digest decision governs retroactively:

- **Confirmation** (explicit, or expiry of the rollback window without objection, per deployment policy) retains the asset in its context region.
- **Rejection** — including rejection *after* the asset has appeared in displayed compositions — SHALL deactivate the asset for all future composition and SHALL trigger cascade treatment for dependents (Annex III §3.5 discipline): pool candidates and staged CDs referencing it are invalidated and regenerated without it; Edition items are re-rendered at next assembly. Rejected asset types dampen future generation of that type (Annex III §5.6 meta-feedback — the operator trains the generator).
- **Audit retention.** A rejected asset is deactivated, not erased: it remains hash-bound in the derivation record so that previously displayed artifacts remain exactly reproducible (VIII.11.3). Deactivation severs future use; it never falsifies history.

VIII.5.6.4 Authority parity and interior constraint. Derived Composition Assets receive no elevated authority from being locally produced: they are subject to the same grammar validation, domain presentation policy (where a governing PCL applies to the composition), per-element provenance display (VIII.4.3, origin class `derived`), and equivalence-class discipline of the Refinement Vocabulary (VIII.6.5.3) as any acquired asset. Charts among them are pre-composed

presentation artifacts: their data values SHALL bind from the declarative and episodic dimensions with ordinary provenance; a Derived chart asserts no ground truth of its own (Suggest-Mode symmetry, Annex IV §IV.3.3).

Informative. The mechanism completes the library model's independence: with only host assets, the system composes plainly; with Derived Composition Assets, it composes richly from its own overnight work; with an admitted PCL, it composes at publisher-curated domain quality. Each stage is a strict superset in capability and identical in governance — and the deployment is never dependent on any publisher to present its own knowledge well.

VIII.6 Prepared Composition Sources and Live Presentation

Prepared visual content reaches the Presentation Surface from three sources, in descending order of preparation specificity: **Candidate Sets** (event-bound, Branch-Condition-matched — VIII.6.1), the **Session Composition Pool** (session-bound, fitness-matched — VIII.6.5), and **staged local generation** as the controlled fallback (VIII.6.6).

VIII.6.1 Candidate Set Generation (Normative)

Visual composition candidates are a candidate-artifact kind within Preparation Run stage 4 (Annex VI §VI.3.2). For a Declared Event, the Preparation Run SHOULD explore the anticipated outcome space by structured multi-branch outcome exploration (informative example: tree-of-thoughts-class methods, applied at taxonomy level where external) and emit a **Candidate Set**: parameterized CDs, each bound to a Branch Condition (e.g. *on time / delayed < 1 h / delayed ≥ 1 h / cancelled*).

Requirements:

1. Every candidate SHALL be fully validated (grammar and domain presentation policy) at preparation time; no validation work beyond slot-level checks remains on the presentation path.
2. Branch Conditions SHALL be evaluable deterministically from locally available runtime data, without inference — consistent with the deterministic-activation discipline of Annex VI §VI.5.1.
3. Candidates SHALL declare their slots such that presentation-time instantiation is pure binding (VIII.4.4).
4. The Candidate Set inherits the provenance and confidence metadata of VIII.4.2 into every candidate.
5. Candidate Sets assembled from admitted content are prepared-package generation (Class 2, Annex III §4.5); candidates requiring new Sources or external artifacts follow the ordinary Class-3 proposal path in the Preparation Brief (Annex VI §VI.3.2, stage 3–4). Sealing, version-stamping, lease-binding, and provisioning of Candidate Sets to devices follow Annex VI §VI.3.2 stage 5 unchanged.

VIII.6.2 Event Session Presentation Path (Normative)

When an Event Session activates (deterministic triggers per Annex VI §VI.5.1), the presentation path for prepared visual content is:

1. **Branch match** — evaluate Branch Conditions against current data (deterministic);

2. **Pool match** — where no Branch Condition matches, fitness-scored retrieval from the Session Composition Pool (VIII.6.5.2);
3. **Selection** — where more than one candidate remains, bounded selection (VIII.6.3);
4. **Instantiation and refinement** — bind current values into the selected candidate's slots; apply Refinement Vocabulary operations where required (VIII.6.5.3);
5. **Render** — deterministic composition by the CE on the Presentation Surface.

This is the prepared-first resolution of Annex VI §VI.5(3) applied to visual artifacts.

Implementations **SHOULD** achieve sub-second time-to-first-display on the branch-match path and sub-second time-to-first-display on the pool-match path, and **SHALL** record, as preparation telemetry in the sense of Annex VI §VI.6(2): **branch hit rate, pool hit rate and fitness distribution**, time-to-first-display, and time-to-complete-composition. These metrics feed the Debrief and the Night Pipeline's iteration unchanged (VI.6(3)); a low hit rate is a preparation defect, not a presentation defect — *the better and broader the candidate preparation, the shorter and more deterministic the presentation path*.

VIII.6.3 Bounded Selection (Normative)

Where the local model participates at presentation time, its task **SHALL** be limited to classification over the prepared Candidate Set: input is the runtime context, output is a candidate identifier and binding map, grammar-constrained. The local model **SHALL NOT** author, modify, or extend candidate content on this path.

VIII.6.4 Instantiation Semantics Across Sources (Normative)

Presentation-time instantiation is pure binding (VIII.4.4) regardless of source. No presentation path authors new composition structure; the paths differ only in how a pre-validated structure is *selected* — by Branch Condition, by Fitness Function, or, in the fallback, by grammar-constrained generation during the attention window.

VIII.6.5 The Session Composition Pool (Normative)

The Session Composition Pool serves the live-interaction case that neither the Prepared Edition nor event-bound Candidate Sets cover: an active session in which information needs arise continuously and unpredictably within a resolvable topic space, and each need is to be answered with a visually high-quality composition at near-zero perceived latency. The mechanism trades cheap local pre-generation in volume for expensive per-need generation: because a candidate is only layout, slot declarations, and hash references into admitted libraries — never pixels — pre-generating hundreds of candidates is inexpensive, and the presentation-time work collapses to deterministic scoring, substitution, and binding.

VIII.6.5.1 Pool Generation. Upon session activation — topic resolution per Annex VI §VI.5(1) or context resolution per Annex II §II.2.2 — and continuously during idle and low-load moments of the session, the local model **MAY** generate pool candidates: parameterized CDs covering the session's topic space, grammar-constrained (VIII.4.1), validated against grammar and domain presentation policy at generation time, and drawing exclusively on admitted content (host assets, admitted PCLs, local data bindings). Pool generation **SHALL** run under an explicit local budget (compute, memory, candidate count) and **SHALL** yield to foreground inference. Where a Preparation Run precedes the session (a Declared Event), the Candidate Set **MAY** seed the pool; pool generation then extends coverage around it. Pool candidates inherit the session's trigger

provenance and preparation lineage (VIII.4.2). The pool is session-scoped: sealed to the session, never shared across trust boundaries, and discarded at session end except as provided in VIII.6.5.4.

VIII.6.5.2 Fitness-Scored Matching. At each information need, a **Fitness Function** SHALL rank pool candidates against the current need. The Fitness Function SHALL be deterministic and reproducible: given identical inputs — the need descriptor, candidate metadata, and pinned scoring parameters (including any embedding model, identified by version and hash) — it SHALL produce identical rankings. The need descriptor MAY be produced by the local model as a bounded task under VIII.6.3 semantics (output: a structured descriptor, grammar-constrained); the scoring itself is algorithmic, never generative. Each match yields a candidate reference and a fitness value (informative example: *candidate #32, fitness 0.90 — substitute element E with asset xyz, bind this text into slots s1, s3*). A deployment SHALL define a minimum fitness threshold; a match below threshold is a pool miss and transitions to the staged fallback (VIII.6.6). A candidate SHALL NOT be presented with a fitness below threshold merely because it is the best available — a mediocre match presented with full visual authority is the "elegant but stale" defect of VIII.7(4) in another form.

VIII.6.5.3 Deterministic Refinement. The residual gap between the matched candidate and the exact need is closed at presentation time exclusively through the **Refinement Vocabulary**: the closed set of transformation operations the governing Composition Grammar declares as permitted on pre-validated candidates. The initial vocabulary comprises: substitution of an element by another element of the same declared equivalence class (same grammar slot type, same geometry class, admitted origin); slot rebinding and reformatting per domain presentation policy; omission of elements declared optional; and re-execution of the constraint solver over the fixed grid. Refinement operations are deterministic, complete in milliseconds, and preserve validation: a refined candidate SHALL revalidate (slot-level and policy checks) before display. Operations outside the vocabulary — structural changes, new elements, free re-layout — are not refinement; they require generation and belong to the fallback path. The vocabulary is registered in the Composition Grammar and grows additively; "arbitrary transformation" is not and never becomes a vocabulary entry (mirroring the register discipline of Annex VII §10.4).

VIII.6.5.4 Pool Lifecycle and Learning. Pool candidates are ephemeral: derived, reconstructible, introducing no new knowledge — their generation, refresh, scoring, and discard are Class-1 operations (VIII.10). Per-candidate usage telemetry (matches, fitness values, operator engagement) is recorded as ordinary evidence. Where telemetry shows a pool candidate pattern recurring across sessions, the Night Pipeline MAY propose its promotion to a persistent, reusable composition template — a new procedural entity, Class 3, proposed in the Morning Digest with the recurrence evidence as its chain, consistent with the recurring-pattern discipline of Annex III §5.7. Promotion is never automatic; an unpromoted pool dies with its session.

Informative — relation to the other sources. The three sources form one coverage gradient: Candidate Sets cover what was specifically anticipated (highest fidelity, branch-deterministic), the pool covers the session's topic space in breadth (high fidelity, fitness-matched, refinement-closed), and staged generation covers the remainder (bounded quality, attention-window-masked). The operator-visible property is uniform: a validated, provenance-complete composition with no perceptible wait — the sources differ only in how much of the work was already done when the need arose.

VIII.6.6 Prepared-Content Miss — Staged Fallback (Normative)

Where no Branch Condition matches and no pool candidate reaches the fitness threshold, the divergence rule of Annex VI §VI.5(4) applies: the absence of prepared content is surfaced as such, nothing is fabricated from stale preparation, and the divergence is recorded for the Debrief. Local full generation of a CD is the controlled fallback, presented in three stages:

1. **Immediate** (target < 100 ms): the textual core of the information plus every already-available prepared element (template skeleton, admitted static assets, media reference links) — with no inference dependency;
2. **Attention window**: grammar-constrained local generation proceeds while the operator reads Stage 1; elements are revealed as they complete validation;
3. **Complete**.

The Stage-1 text is content, not placeholder: its reading time is the budget within which Stage 2 completes. On timeout, affected slots degrade per VIII.7(4).

Informative — the governing property. Perceived real-time is achieved by construction, not by speed alone: on a branch hit, because nothing expensive remains to compute; on a pool hit, because scoring, substitution, and binding are millisecond-scale deterministic operations; on a full miss, because the waiting interval is absorbed by the staged textual core. A presentation is real-time conformant if at no point an operator-perceptible waiting interval exists between the moment information is needed and the moment a correct, validated representation of it is available — beginning with its textual core.

VIII.7 Presentation Integrity (Normative, All Paths)

1. **Validation before display.** No element SHALL be displayed before its CD region has passed grammar validation and domain-presentation-policy checks.
 2. **No layout shift.** Late-arriving elements fill space reserved by fixed slot geometry; they SHALL NOT displace, resize, or reflow content already displayed.
 3. **Atomic reveal.** An element appears once, complete and final; displayed content SHALL NOT be revised by continued generation. Later invalidation SHALL be presented as an explicit update, never a silent replacement (no-silent-error principle).
 4. **Degradation chain.** Every composed artifact SHALL define a descending chain — full composition → reduced template of the same grammar family → tabular representation → textual representation — triggered by prepared-content miss without viable fallback result (VIII.6.6), validation or revalidation failure (including failed refinement revalidation, VIII.6.5.3), sub-threshold fitness, budget exhaustion, or Stage-2 timeout. Degradation SHALL be silent and complete, per the graceful-degradation discipline of Annex II §II.2.8: a plain table is a fulfilled answer; a progress indicator is a promise. The system SHALL NOT display a high-fidelity composition whose bound data no longer matches its prepared candidate; it SHALL degrade instead.
-

VIII.8 The Prepared Edition

VIII.8.1 Position (Normative)

The Prepared Edition is the visual presentation form of the Morning Digest and of Preparation Briefs. It introduces no new governance object: every item it renders is a Digest or Brief item in the full sense of Annex III §5.6 and Annex VI §VI.3.3, with its evidence link, provenance, and mutation class; every Class-3 item remains a proposal with explicit accept / reject / defer affordances, each decision a Consent Record; the Digest's hard budget and meta-feedback rules apply unchanged. The Edition changes how this content is *presented* — editorial page form, large uncluttered infographics for quick ingest, priority-driven layout — never what it is.

The Edition is assembled entirely within the preparation window from admitted content and validated CDs; at presentation, only slot refresh and deterministic rendering occur. The Edition SHALL exhibit no operator-perceptible composition latency.

Informative — form. Editorial page grammar is an established, universally readable system for encoding importance as visual salience — lead placement, column width, headline scale, image size — and is adopted here rather than invented.

VIII.8.2 Priority-to-Salience Mapping (Normative)

Edition layout SHALL be derived from the local ranking of the Digest and Brief content (expected utility × evidence strength, Annex III §5.6) by a deterministic mapping from ranking weight to layout weight, resolved against the governing grid grammar by constraint solving. The reasoning that produced the ranking decides *what matters and what belongs together*; the grid decides *where and how large*. The mapping function and the grid SHALL be local.

VIII.8.3 Sections by Trigger Provenance (Normative)

Content originating from Forward Preparation (**declared** — Brief items of confirmed Declared Events) and from Predictive Forward Preparation (**inferred** — Preparation Queue entries, Annex VI §VI.4) SHALL be visually distinguishable at all times and SHALL NOT share visual authority:

- **Declared content** occupies the editorial sections: lead-position eligibility, full salience range, authoritative presentation.
- **Inferred content** SHALL be confined to a recognizably distinct advisory section with a capped salience ceiling below the declared range. Confidence MAY modulate ordering within the advisory section but SHALL NOT raise an inferred item into the declared salience range.

Because no retrieval, generation, or provisioning occurs for a predicted event before acceptance (Annex VI §VI.4), inferred Edition items are renderings of Queue entries — what would be prepared, why (evidence chain), and at what cost — composed exclusively from locally held material. Accepting an inferred item on the Edition surface is the Queue-acceptance consent of VI.4 (converting the entry into a Declared Event with scoped Preparation Consent), rendered in place; it is never implicit in viewing, dwelling, or expanding.

Rationale (informative): inferred content is a system hypothesis. Presenting hypotheses with the visual authority of confirmed preparation lends speculation borrowed credibility — the mechanism

by which personalization becomes manipulation. Binding trigger provenance to salience closes this channel structurally, and matches the corpus discipline that proposals are visibly proposals.

VIII.8.4 Multi-Publisher Pages (Normative)

Where content associated with more than one publisher appears on one Edition page:

1. **Ranking is local.** Cross-publisher ordering and salience SHALL derive exclusively from locally recorded, policy-authorized signals, per the routing-state exclusivity of Annex II §II.2.5 as refined by Annex IV §IV.5.8. No PCL mechanism, manifest field, or policy SHALL be capable of influencing its content's rank or prominence relative to other publishers; offending declarations SHALL be ignored and the event logged.
2. **The host theme governs.** Page-level visual coherence is provided by a local host theme; PCL elements are embedded within it. Publisher themes apply, at most, within the interior of that publisher's own elements.
3. **Attribution is per element.** Precisely because the page reads as one coherent artifact, per-element provenance (VIII.4.3) SHALL remain visually resolvable — unobtrusive but discoverable, consistent with Annex V §V.4(2).

VIII.8.5 Deep-Dive Descent (Normative)

Every Edition item SHALL be expandable per the Deep-Dive Expansion model of Annex VI §VI.3.4, unchanged: **inspection** (immediate — full content, evidence chain, provenance, dependencies; no new computation, no new retrieval) and **focused follow-up preparation** (on request — a scoped Preparation Run under the existing Preparation Consent, or a new narrower consent where the sub-point exceeds the original scope). Deep-dive enrichment prepared during the preparation window — including material prepared in support of adapter generation (Annex III §6.1, stages 4–6) — SHALL be linked from the corresponding Edition items as inspection-depth targets. Descent at inspection depth SHALL NOT trigger external retrieval. Expansion never bypasses a class boundary: deepening a proposal does not accept it (VI.3.4).

VIII.8.6 Assembly Budget (Normative)

Edition assembly for inferred content is bounded by the Digest's hard budget (Annex III §5.6) and SHALL favor precision over recall — the dampening discipline of §5.6 and VI.6(3) applies: item types repeatedly rendered but ignored are dampened. Declared preparation is not budget-capped by this section; it is owed completeness within its consented scope. Full-fidelity, PCL-governed composition is warranted primarily for declared content; inferred items MAY be deliberately plainer, which simultaneously respects the budget and the salience ceiling of VIII.8.3.

VIII.9 The Augmented Overlay

VIII.9.1 Position and Toggle Discipline (Normative)

The Augmented Overlay is an interactive layer rendered *over* composed artifacts and Presentation Surfaces — annotations anchored to displayed elements, context-bound automation suggestions, and spoken and cursor-based interaction affordances. It is the specified instance of the interactivity rule of VIII.4.5(1): the composed artifact remains strictly passive; everything interactive lives in the overlay layer provided by the runtime.

The overlay is **operator-toggable at all times**: globally, per Presentation Surface, and per context region. Toggle state is local configuration; no publisher mechanism, PCL declaration, grant, or delivered content SHALL be capable of activating the overlay, preventing its deactivation, or conditioning any content or function on its being active. Every function of the system — reading, Deep-Dive descent, consent decisions, prepared-action confirmation — SHALL remain fully available with the overlay off; the overlay adds affordances, it never gates them.

Informative. A permanently active suggestion layer is an attention defect, not a feature. The toggle discipline treats overlay attention as an operator resource in exactly the sense the Digest budget treats consent attention (Annex III §5.6): spent only where the operator chooses, and trainable — ignored overlay content dampens (VIII.9.4).

VIII.9.2 Overlay Content — Composed, Region-Bound (Normative)

Overlay content is composed content: annotation and suggestion elements are specified by CDs, validated against the Composition Grammar, rendered by the CE as strictly passive HTML/CSS per VIII.4.5, and subject to per-element provenance (VIII.4.3) unchanged. Overlay content originates from the same two library provenance classes:

1. **Acquired.** A publisher MAY ship overlay content in its PCL — annotation sets, guidance layers, and automation templates for its domain. Overlay automation templates are session/execution templates in the ordinary sense of Annex IV §IV.6.2: admitted Class-3 (Annex IV §IV.6.4), verification-chained (Annex VII §7), and — decisively — **region-bound at admission**: publisher overlay content SHALL be bound to the context regions of the publisher's declared scope and SHALL NOT surface outside them, extending the region-binding discipline of Annex IV §IV.3.4 from parametric competence to overlay content. An automation is specific to its context; its suggestion surface is bound to the same address space. Distribution and hosting of overlay content follow VIII.5.1 without exception: overlay content of public offerings is hosted, publicly auditable, in the public Optirando™ repository under the publisher's Manifest Root; non-public overlay content follows the secret path, hash-committed under the same root.
2. **Derived.** The system MAY prepare overlay content for a context region itself: Forward Preparation stages event-scoped overlay material under the Preparation Consent (annotations and automation suggestions for the Declared Event's regions, per Annex VI §VI.3.2 stage 4), and Predictive Forward Preparation MAY propose overlay preparation via the Preparation Queue (Annex VI §VI.4 — nothing generated before acceptance). Derived overlay annotations follow VIII.5.6 unchanged (Class-2 staging, Digest reporting, retroactive rejection); derived automation paths are procedural entities and follow the WCR and skill lifecycle of Annexes II–III unchanged (Class 3 where new procedural capability is created).

All overlay content — both classes — is anchored to context regions, never global: surfacing is a deterministic function of the resolved region and the overlay toggle state.

VIII.9.3 Automation Suggestions and the Execution Boundary (Normative)

Overlay automation suggestions are the presentation surface of the existing two-stage skill routing (Annex III §6.4): semantic candidate retrieval, then precondition filtering — target reachability, region match, authorization state. Below the confidence threshold a matching automation is *offered*,

never staged; above it, it MAY enter the standard execution preview. Nothing changes at the boundary:

- A suggestion — however tailored, however prepared — pre-satisfies nothing (Annex II §II.2.5). Every consequential step remains gated at its applicable PoAE™ tier at the point of consequence.
- Publisher-delivered prepared actions surfacing in the overlay remain prepared actions under their standing scope (Annex VII §10.3): pinned template, schema-validated payload, effect displayed precisely, one consent tap, PoAE™ record. The overlay changes where they render, never what they are.
- Suggestion ordering and selection derive exclusively from locally recorded, policy-authorized signals (Annex II §II.2.5; Annex IV §IV.5.8); no publisher mechanism influences the prominence of its own suggestions relative to others (VIII.8.4(1) applied to the overlay).

VIII.9.4 Interaction Modalities and Evidence (Normative)

Spoken interaction SHALL be transcribed exclusively by the local speech pipeline (the Whisper-class discipline of Annex VI §VI.3.1); no utterance leaves the deployment. Spoken input MAY navigate, query, annotate, and trigger suggestions; it SHALL NOT substitute for any PoAE™ authorization factor — a spoken "yes" is input, not proof.

Cursor and gaze-free pointer interaction with overlay elements (hover, select, dismiss, expand) is local interaction evidence. All overlay interaction telemetry — what was engaged, ignored, dismissed — is recorded exclusively as local evidence events (Annex III §5.2 pattern) and SHALL NOT be observable by, or derivable to, any publisher (routing-state exclusivity; the retrieval-privacy threat model of Annex VI §VI.7.1 applies to interaction patterns identically). Dismissed and ignored overlay content types dampen future surfacing (Annex III §5.6 meta-feedback).

Topic triggering. The overlay MAY offer, on any displayed element or region, the operator affordances "*prepare this*" and "*deepen this*": the former creates a Declared Event via the ordinary declaration path (Annex VI §VI.3.1 — parsed descriptor, confirmation, scoped Preparation Consent); the latter marks the topic as an evidence-backed gap candidate for the Night Pipeline, feeding the ordinary gap typology and escalation ladder (Annex III §§5.4, 6.2) — up to and including, where the ladder justifies it, data-set generation and adapter training as Class-3 proposals in the Morning Digest. The overlay is thereby a capture surface for the learning loop; it holds no shortcut into it: triggering creates evidence and proposals, never trained artifacts.

VIII.9.5 Classification (Normative)

Overlay surfacing decisions (region-resolved candidate selection, re-ranking against interaction evidence) are Class 1–2 per the existing assignment table (skill re-ranking for context: Class 2). Admission of publisher overlay content: Class 3 (Annex IV §IV.6.4). Derived overlay annotations: per VIII.5.6.3. New derived automation paths: Class 3. Overlay interaction evidence recording: Class 1. The toggle state itself is local configuration; changing it is not a graph mutation.

VIII.10 PoAC™ Classification (Normative)

Per the extension rule of Annex III §4.5, the assignment table is extended:

Operation	Class
CD rendering by the CE from a validated CD and admitted assets	1
Session Composition Pool generation, refresh, fitness scoring, refinement, and discard (ephemeral, reconstructible, admitted content only)	1
Presentation telemetry recording (branch hit rate, pool hit rate, fitness distribution, display timings)	1
Overlay interaction evidence recording (engagement, dismissal, ignore telemetry)	1
Overlay surfacing and re-ranking of admitted content within resolved regions	2
Candidate Set / Edition assembly from admitted content (prepared-package generation)	2
Generation and staging of Derived Composition Assets from admitted content (reported item-by-item in the Morning Digest, one-action rollback with cascade)	2
Retroactive rejection of a Derived Composition Asset (deactivation + dependent cascade, derivation record retained)	2
Promotion of a recurring pool candidate to a persistent composition template	3
Candidate re-ranking, salience-mapping parameter adjustment within admitted grammars	2
PCL acquisition and admission (whole or per artifact)	3
Admission of publisher overlay content and automation templates (per Annex IV §IV.6.4)	3
Creation of a new derived automation path surfaced via the overlay	3
External reasoning interaction for composition candidates (taxonomy-level, under Preparation Consent)	3, or covered by the declared scope of the Preparation Consent per Annex VI §VI.3.1
Ingestion of any new Source surfaced by composition preparation	3
Domain-presentation-policy configuration change (local overrides)	3
Nothing in this table weakens Annex III §4.4: no new knowledge or capability enters the WR Graph™ without a Consent Record.	

VIII.11 Governance and Privacy Invariants

VIII.11.1 Grant Model and Execution Boundary (Normative)

All activity under this Annex is preparation, delivery, or presentation. Under Annex VII §10: PCL delivery and update ride on delivery/preparation rights; standing scopes never bundle (VII-10.5.1) and never expand through composition machinery. Presentation of a composed artifact — including one that renders a prepared action or recommends an act — confers no authority: prepared actions remain prepared actions, executed only by the human consent tap with a PoAE™ record (VII-10.1), and no CD, candidate, salience level, or Edition placement pre-satisfies any PoAE™ tier (Annex II §II.2.5).

Predictive Forward Preparation remains bounded exactly as in Annex VI §VI.4: no external retrieval, no package generation, no provisioning before acceptance. Composition for inferred items operates exclusively on locally held material (VIII.8.3). A system inference SHALL NOT be grounds for publisher requests, scope requests, or acquisitions; where an accepted item's preparation requires material beyond its consented scope, the ordinary scope-extension consent of VI.3.4 applies.

VIII.11.2 Retrieval Privacy (Normative)

All external interaction in service of composition preparation is subject to Annex VI §VI.7 in full: masking baseline always; Obfuscation Mode per policy, mandatory in High-Assurance Mode; Query Distribution where active; local-only preparation permanently available as the conforming fallback. Presentation-time media retrieval is additionally constrained per VIII.5.5: it SHALL NOT be correlatable by the publisher with Event Session occurrence or preparation content.

Structural consequence (informative): the publisher supplies attested assets and composition grammar without learning which of it is used, when, or why. Personalization of the Edition exists entirely on the local side, derived from locally recorded signals — never from publisher input — consistent with the routing-state exclusivity that governs the rest of the corpus.

VIII.11.3 Auditability (Normative)

For every displayed composed artifact, the system SHALL retain: the CD (and, where applicable, the Candidate Set and the matched Branch Condition), the Composition Grammar version it validated against, the hash-bound asset set, the domain-presentation-policy version applied, and the preparation lineage reference (Preparation Consent / Queue entry). Given these, the artifact SHALL be exactly reproducible. This is the presentation-layer instance of the corpus's proof discipline: what was shown, from what, under which rules, is demonstrable after the fact.

VIII.12 Relation to Prior Work and Claimed Contribution (Informative)

The constituent mechanisms are established practice and are not claimed as novel: generation of declarative layout and visualization specifications by language models with deterministic downstream rendering; grammar- and schema-constrained decoding; declarative visualization grammars and constraint-based layout; precomputation and amortization of expensive work outside latency-critical paths (of which real-time graphics reconstruction is the canonical example); and perceived-latency techniques — staged display, layout stability, graceful degradation.

The claimed contribution is the normative composition under the corpus's governance model: the placement of neural computation as a conformance invariant with a deterministic, reproducible presentation path (the deliberate inversion of the graphics precedent, required because information — unlike pixels — carries correctness obligations); composition-not-synthesis with per-element provenance over exclusively provenance-bound building blocks, with asset synthesis confined to the governed, reversible preparation phase; the composition library in two provenance classes — the Publisher Composition Library as a verification-chained, consent-admitted, grant-delivered trust object whose grammars bound the local composition space, and Derived Composition Assets as regenerable, Digest-reported, retroactively rejectable local counterparts — making the deployment

independent of any publisher for presenting its own knowledge; branched candidate preparation and the session-scoped composition pool — cheap local pre-generation in volume, deterministic fitness-scored retrieval, and a closed refinement vocabulary — binding presentation latency to preparation quality, with their telemetry closed into the Debrief loop; the normative binding of trigger provenance and confidence to visual salience, so that system hypotheses can never borrow the visual authority of operator-confirmed preparation; the Augmented Overlay as a strictly optional, region-bound, operator-toggleable interaction layer whose suggestions surface the existing routing, preparation, and learning machinery without holding any shortcut into execution or training; and the exclusion, by construction, of any publisher influence on cross-publisher ranking and of any presentation-layer path to execution authority.

§VIII.13 Version History

Version	Date	Change
1.0	2026-07-11	Initial normative specification: hybrid compute split with Cloud Abstraction Boundary compliance and mandatory local instantiation; Inversion Rule and composition-not-synthesis rule; Composition Description with mandatory trigger-provenance/confidence metadata, per-element provenance, fixed-geometry slots, and the HTML/CSS rendering target (strict passivity — no script or active content; library-only, hash-addressed resource resolution with zero render-time network activity; pinned deterministic feature subset; exact self-containment); Composition Libraries in two provenance classes: the Publisher Composition Library under Annex VII artifact machinery with Class-3 admission, Annex V reconstruction of media-class assets, layout grammars, exemplars, domain presentation policy, reference-only media integration, public-repository hosting of all public-offering library and overlay content under the Manifest Root with full public auditability per Annex IV §IV.6.3(1), secret-path hash commitment under the same per-provider root per Annex VII §13.2, and mandatory publication-side reconstruction (no media originals distributable or resolvable; encrypted, hash-bound repository-side custody retention of originals for evidence per Annex V custody discipline; repository-side tool-governed Trusted Converter; WebP as the pinned raster serialization for library imagery, analogous to safetensors for adapters; secret-path media reconstructed by the receiving deployment's sandbox pipeline per Annex V; additive to, never a substitute for, receiving-side Annex V obligations); and Derived Composition Assets — Night-Pipeline-generated icons, graphics, and pre-composed charts anchored to topic context regions, tool-governed generation, full derivation chains with regenerability, Class-2 staging with Morning Digest item-by-item reporting, one-action rollback, retroactive rejection with dependent cascade and audit-preserving deactivation, and authority parity with acquired assets; branched candidate preparation as a Preparation-Run stage-4 artifact kind with deterministic Branch Conditions and bounded selection; the Session Composition Pool — session-scoped, budgeted local pre-generation of validated candidates over the session's topic space, deterministic fitness-scored matching with a mandatory minimum threshold, a closed deterministic Refinement Vocabulary (equivalence-class substitution, slot rebinding, optional-element omission, constraint re-solve) with mandatory revalidation, Class-1 ephemeral lifecycle with Class-3 promotion of recurring candidates via the Night Pipeline; staged prepared-content-miss fallback and telemetry (branch and pool hit rates, fitness distribution) into the Debrief; presentation integrity rules (validation before display, no layout shift, atomic reveal, degradation chain per Annex II §II.2.8); the

Version Date**Change**

Prepared Edition as the visual presentation form of the Morning Digest and Preparation Briefs with priority-to-salience mapping, trigger-provenance salience binding, local-only cross-publisher ranking, per-element attribution, Deep-Dive descent per Annex VI §VI.3.4, and Digest-budget discipline; the Augmented Overlay — operator-toggleable at all times (globally, per surface, per region) with structural inability of publishers to activate it or condition function on it, strictly passive composed overlay content from both library classes with mandatory region binding at admission, automation suggestions as the presentation surface of two-stage skill routing with the execution boundary unchanged, local-only speech transcription and interaction evidence with meta-feedback dampening, and "prepare this" / "deepen this" affordances feeding the declaration path and the gap/escalation machinery including adapter-generation proposals; PoAC™ assignment-table extension; governance and privacy invariants (execution boundary, predictive bounds, retrieval privacy, exact reproducibility).

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End of Annex VIII.